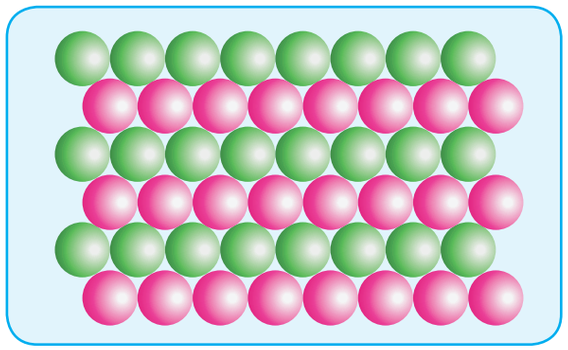
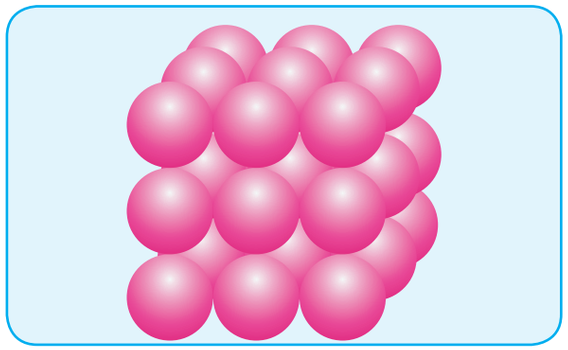
(i) ABAB.. Type:

In this type, the second row spheres are arranged in such a way that they fit in the depression of the first row as shown in the figure. The second row is denoted as B type. The third row is arranged similar to the first row A, and the fourth one is arranged similar to second one. i.e., the pattern is repeated as ABAB….In this arrangement each sphere is in contact with 6 of its neighbouring spheres.



On comparing these two arrangements (AAAA...type and ABAB….type) we found that the closest arrangement is ABAB…type.



Simple Cubic (SC)

6.6.3 Simple cubic arrangement:

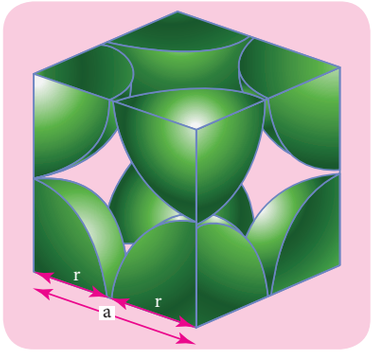
This type of three dimensional packing arrangements can be obtained by repeating the AAAA type two dimensional arrangements in three dimensions. i.e., spheres in one layer sitting directly on the top of those in the previous layer so that all layers are identical. All spheres of different layers of crystal are perfectly aligned horizontally and also vertically, so that any unit cell of such arrangement as simple cubic structure as shown in fig.

In simple cubic packing, each sphere is in contact with 6 neighbouring spheres - Four in its own layer, one above and one below and hence the coordination number of the sphere in simple cubic arrangement is 6.

Packing efficiency:

There is some free space between the spheres of a single layer and the spheres of successive layers. The percentage of total volume occupied by these constituent spheres gives the packing efficiency of an arrangement. Let us calculate the packing efficiency in simple cubic arrangement,

$$\left\{\begin{matrix}\text{Packing fraction}\\ \text{(or) efficiency}\end{matrix}\right\} = \frac{\left\{\begin{matrix}\text{Total volume occupied by}\\ \text{spheres in a unit cell}\end{matrix}\right\}}{\text{Volume of the unit cell}} \times 100$$



Let us consider a cube with an edge length 'a' as shown in fig.

Volume of the cube with edge length a is = a × a × a = a^3

Let 'r' is the radius of the sphere. From the figure, a=2r $\Rightarrow$ $r = \frac{a}{2}$